

Quality-Normalized AI Compression

Measuring speed only after scope, complexity, quality, evidence, and recovery are held constant

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Executive summary

AI productivity claims are easy to inflate when faster work is compared with a weaker result. This paper proposes quality-normalized compression: compare time or effort only after establishing equivalent scope, complexity, quality, evidence, and recovery requirements.

Why raw speed is a weak metric

A faster result is not automatically a better production system. If the faster path drops requirements, reduces QA, skips recovery, or produces weaker evidence, the comparison is not like-for-like. Enterprise AI needs a measurement discipline that treats quality as a constraint rather than an optional adjustment.

The normalization rule

A defensible compression claim requires an equal-output contract. At minimum, comparisons should normalize:

- scope and requirement denominator;
- task complexity and dependency burden;
- quality/acceptance criteria;
- evidence and auditability;
- recovery/rollback obligations;
- lifecycle target;
- human and system boundary conditions.

If the AI-assisted route is faster only because one of these is weaker, the correct conclusion is **faster but weaker**, not genuine compression.

Measured, documented, modeled, target

Metrics should carry evidence classes. Measured time comes from a trustworthy event or timer source. Documented time is recorded in a source but not independently instrumented. Modeled time estimates a counterfactual. A target is an aspiration. These categories should not be silently merged into one number.

Three-lane comparison

A useful enterprise comparison can separate three lanes:

Lane	Meaning
Traditional no-AI	Human workflow without generative/agentic assistance
AI-assisted no governed system	AI used as a productivity tool without the full assurance architecture

Lane	Meaning
Governed AI execution	AI-assisted work with equivalent scope, QA, evidence, recovery, and lifecycle controls

The purpose is not to guarantee that one lane always wins. It is to keep the comparison honest.

Wall time and person-hours are different

AI systems can compress active human effort without compressing wall-clock time, or vice versa. Parallelism can reduce elapsed time while consuming substantial compute. Reviews and owner gates can add waiting time without adding person-hours. A credible metric package therefore separates active time, wall time, owner effort, system execution, QA/rework, and external waiting.

Cost is also multi-layered

Actual spend, replacement cost, and modeled savings answer different questions. Actual spend is what was paid. Replacement cost estimates what an equivalent alternative might cost. Savings are a derived comparison and depend on assumptions. Keeping these layers separate prevents a modeled business case from being presented as observed cash savings.

Quality-normalized compression as an enterprise proof

Once equal scope and quality are established, compression can become powerful evidence. It can show that governed AI does not have to trade assurance for speed. But the measurement must preserve the denominator and make uncertainty visible.

A publication checklist for productivity claims

Before publishing a productivity multiplier, disclose the compared scope, quality controls, evidence strength, time basis, cost basis, sample size, and major limitations. Avoid universalizing a project-specific result.

Limitations

Counterfactual labor and cost estimates remain models unless directly observed. Small samples may not generalize. Quality-normalized compression is a measurement discipline, not a guarantee of any specific multiplier or economic outcome.

References

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Source-currentness note

External references [1]-[5] are carried forward from the independently reviewed and publicly released WP-02 v1.0.0 canary. This batch does not claim a new live-web currentness check. Time-sensitive statistics are used only where explicitly cited and dated. First-party references [6]-[7] are BRAIZ Works publications.

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